Technology empowerment in disability employment: a bibliometric and systematic review

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TECHNOLOGY EMPOWERMENT IN DISABILITY EMPLOYMENT: A BIBLIOMETRIC AND SYSTEMATIC REVIEW

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ABSTRACT

Purpose – We examined the role of technology in the employment and work for people with disabilities (PWD). We intended to identify publication trends, key terms, research areas, and potential gaps in the topic.

Design/methodology/approach – We scrutinized relevant articles from two major databases through bibliometric and systematic review analyses. The bibliometric data involved 256 articles analysed through the VOSviewer software to produce publication trends and network maps. We then scanned article abstracts and selected 114 articles for thematic analysis using the PRISMA guideline.

Findings – The findings revealed a dominance of authors from Western and European countries, particularly the United States. Three themes emerged as the crucial factors shaping technology utilization for PWD employment and workplace inclusion. These were identified as individual, environmental, and societal; which shed light on the multifaceted influences surrounding PWD's technology empowerment.

Originality/value – The hybrid review approach enables a thorough examination of the published work. Our study proposes a conceptual model for PWD employment and workplace inclusion. Founded on the theories of Work Adjustment, Human-Computer Interaction, and the Social Model of Disability. The model provides a clear visualization to guide further research and relevant initiatives in this area.

Keywords disability employment, technology empowerment, assistive technology

Paper type Literature review

1. INTRODUCTION

Globally, people with disabilities (PWD) suffer higher unemployment of 7.6 percent average compared to non-disabled peers (Stoevska, 2022). PWD also face significant challenges in employment access, animosity, and discrimination (Lindsay *et al.*, 2023). Strategies at the international (ILO, 2023), regional (European Union, 2021), and country-specific levels (Jamil and Saidin, 2018) suggest commitment to address this issue. Shifting medical to a social disability model (Hogan, 2019; Shakespeare, 2006) in the past few decades marks a critical milestone in disability inclusion. The World Health Organization (WHO) defines disability as a relational concept influenced by individual abilities and characteristics within specific social and developed environments (WHO, 2002). Barriers of inclusion stem from societal and economic processes (WHO, 2022) .

Social technology can empower PWD in all aspect of life, including employment (Nierling and Maia, 2020). Borrowing ideas from the Psychological Empowerment Theory (Zimmerman, 1995; Zimmerman *et al.*, 1992) and the Capability Approach (Nussbaum, 1995), 'technology empowerment' describes the use of technology to assist PWD in navigating their lives. Empowerment occurs when PWDs attain capability and independence. Accordingly, we defined technology empowerment as PWD's enhanced capabilities of using technological tools for employment access, work functionality, and productivity.

Assistive technology (AT) tools are pivotal for achieving this goal (Cazini and Frasson, 2013). AT refers as any piece of technological equipment used by PWD to perform/enhance functionalities in work performance. AT is used in job accommodation to adjust the work environment which enables PWD's to fully participation (Chow *et al.*, 2014). Targeted AT improves PWD's accessibility, well-being, and inclusivity.

Targeted AT Tools include screen readers, augmentative communication devices, adaptive computer input devices, smartphone apps, environmental control systems, and job-specific software. Advanced technology offers better tools to accommodate diverse forms of disabilities. Human-centered innovation allows customized AT experience to suit an array of disabilities. AT potentials are limitless Schwanke et al. (2005) hence, creating prospects for PWD technological empowerment (Wong *et al.*, 2021).

Despite its potential, the extent of AT use remains underexplored. Studies point to problems such as limited accessibility, mismatched needs, and inadequate support (Dong *et al.*, 2022). If well-implemented, technology offers substantial benefits for disabled employees and employers, contributing to work-life expectations (Heinz *et al.*, 2021). These gaps trigger the need to review related studies on the topic.

Previous review studies are available but restricted in some ways. For instance, Lancioni et al. (2021) reviewed literature on specific technologies tailored to particular disabilities. Albeit niche insights are limited in providing a helicopter view regarding the whole topic. Furthermore, these studies focused on wider socio-economic aspects in PWD affairs rather than specifically on employment. Review specific on technology regarding disability employment remains open for exploration. Additionally, the goal of review studies is to identify research trends and gaps, older papers may be outdated (Alper and Raharinirina, 2006).

Hence, our study examined technology empowerment in disability employment through reviews of relevant articles. The specific objectives were:

1. To map publications on technology empowerment in disability employment.
2. To uncover common keywords and research areas on the subject.
3. To determine key factors influencing technology empowerment in disability employment.
4. To suggest gaps and research models for future studies on the subject.

In achieving the objectives, we employed bibliometric analysis (BA) and systematic review (SR) approaches. Objectives 1 and 2 were achieved through BA on two databases. Objective 3 involved SR on selected articles. Whereas objective 4 integrated both giving insights to suggest gaps and research models.

This paper is structured accordingly. The next section explains the methodology, followed by the results. Then, discussion and emerging models are presented. The conclusion highlights the contributions, recommendations, and limitations.

2. METHODS

2.1 Overview

We adopted BA and SR for a robust literature analysis. BA is widely recognized for analysing scholarly outputs to map publication patterns, authors, and keywords (Donthu et al., 2021). For BA, we analysed 256 articles using the VOSviewer (version 1.6.7). From this, we identified 114[1] articles that mentioned terms like workplace, job, employer, or occupation reflecting AT usability at work. SR on the 114 articles followed the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines (Moher *et al.*, 2010). This involved full-text assessment focusing on study samples, design, interventions, and outcomes. The selected articles allowed focus on factors influencing technology empowerment in disability employment. The articles were analysed for relevant codes, sub-themes, and themes to understand the research landscape (Petticrew, 2006). Figure I outlines the study flowchart.

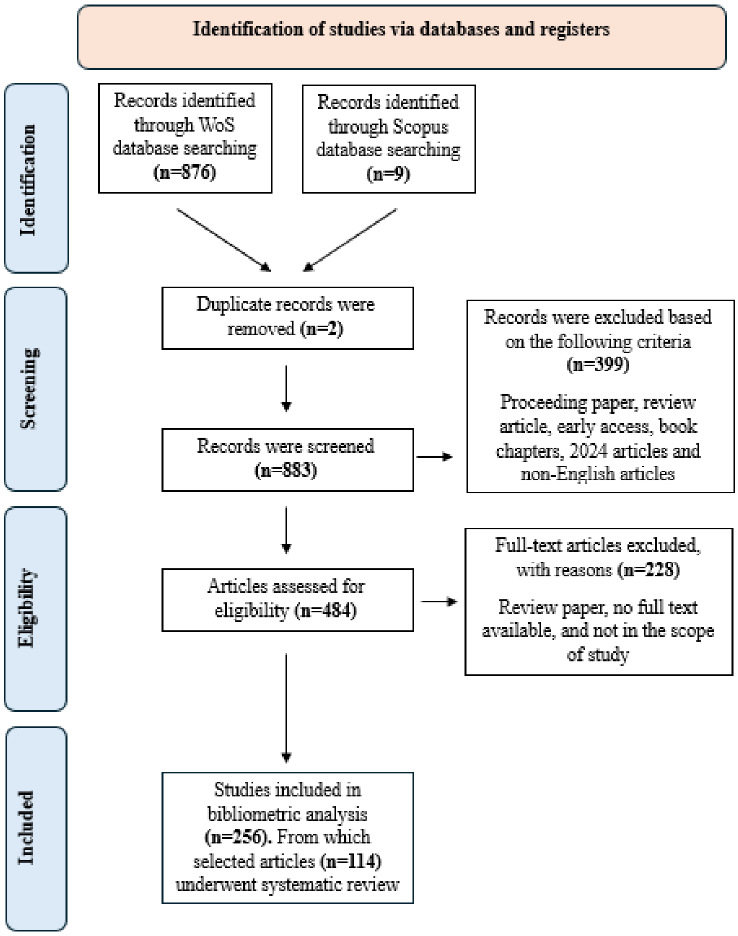

Figure I PRISMA study flowchart

Source: Adapted from (Moher *et al.*, 2010).

2.2 Database selection

The Web of Science (WoS) and Scopus were chosen for their global prominence (Norris and Oppenheim, 2007). Although considerably overlap, each exhibits its strengths. WoS extensively covers natural sciences and engineering, whereas Scopus has higher social science documents (Mongeon and Paul-Hus, 2016). Combining both allows comprehensive data for the study. The initial dataset after removing duplicate documents consisted of 883 records. After screening, the final articles comprise of 256.

2.3 Search strategy

The search commenced on January 20, 2024. We followed three phases to identify articles. The first involved keyword recognition. Search keywords are critical in review studies influencing the number and specificity of generated articles.

Following the study aim, we focused on target group, intervention, and outcome, and identified synonyms based on common language usage (Table *I*), "impairment" and "impaired" followed WHO's disability definition. To reflect technology application at work, 'workplace' and 'accommodation' keywords were used encompassing a range of adjustments, modifications, and supports for creating inclusive environment.

Table *I* Finalized search terms

Population: **PWD**	**Intervention:** **AT and computer**	**Outcome:** **workplace and accommodations**
• Disability • Disabled • Disabilities • Impairment • Impaired	• Assistive technolog(y)(ies) • Computer	• Workplace • Accommodation(s)

Source: Table by authors

For intervention, we deliberately adopted general keywords of "assistive technolog(y)(ies)" and "computer" for several reasons. The AT term is widely utilized in the discourse of disability technology empowerment, particularly relating to work accommodations (Wong *et al.*, 2021). The term encompasses a broad range of tools/devices/software/adaptations designed to enhance PWD functionality and independence. The broad-based search approach agrees with the overarching AT definition by WHO.

Selection of the search terms suits our intention to capture technological solutions for PWD employment. Given diversity of AT tools, applying specific keywords has the risk of overlooking relevant literature. Furthermore, the generic term provides a feasible approach, an important consideration when reviewing several databases.

Table *II* displays the search strings, to search article titles, abstracts, and keywords. The 'author keywords' was used instead of 'keywords plus' to generate focused results (Juan Zhang *et al.*, 2015). The search retrieved 883 papers after removing 2 duplicates.

Table *II* Search strings

Web of Science	(TS=(("assistive technolog*" AND computer* AND disab* OR impair* AND "accommodation*" AND workplace))) NOT KP=((assistive technolog* AND computer* AND disab* OR impair* AND"accommodation*" AND workplace))
Scopus	TITLE-ABS-KEY(" assistive technolog*" AND computer* AND disab* OR impair* AND "accommodation*" AND workplace)

Source: Table by authors

The second phase involved screening 883 papers based on inclusion and exclusion criteria (Table *III*). The focus on research-based articles aligned with the study objectives was to examine trends and gaps. Using these parameters, 399 publications were excluded.

In third phase, known as eligibility, the remaining 484 articles were examined, reviewing titles and abstracts for alignment with the study. From these, 228 papers did not meet the requirements hence excluded as most pertained non-working settings.

Table *III* Selection criteria

Criterion	**Inclusion**	**Exclusion**
Timeline	<2023	2024
Literature Type	Journal (Article)	Review papers, book chapters, proceeding papers, and early access.
Language	English	Non-English

Source: Table by authors

After screening, 256 articles remained for BA. From these, 114 articles were analysed for SR using PRISMA method. The study then proposed a conceptual model based on emerging themes/sub-themes. Integration of BA and SR provided the study with a comprehensive analysis of the topic.

3. RESULTS

3.1 Bibliometric results

i. Publication output

Figure II shows 1994–2023 publication trends. The articles increased by 5–10 annually between 1994 and 2007. In 2008, 10–20 publications increased due to technological advancements in addressing various disability types. The publications decreased in 2019, possibly disrupted by the COVID-19 pandemic. The trend increased after 2020, suggesting accelerated interests in this area.

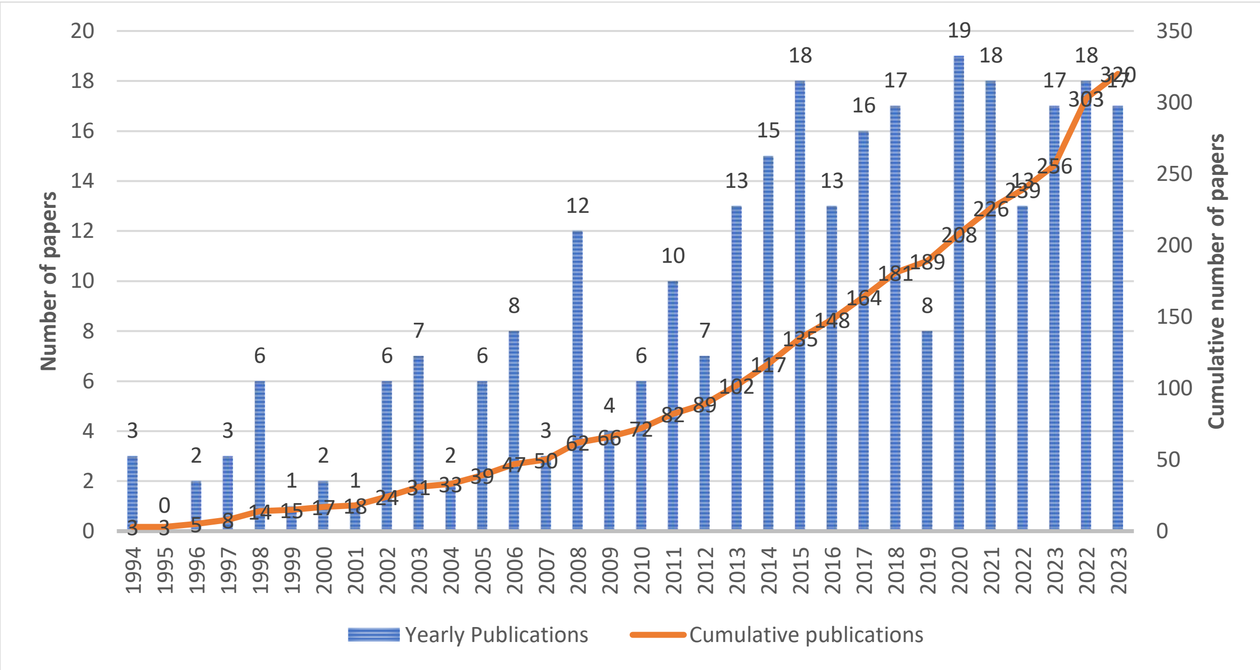

Figure II The publication trends 1994-2023

Source: Figure by authors

ii. Top 10 most productive countries

Table *IV* indicates dominance of certain Western and European countries, fronted by the United States (US). Among Asian countries, China led followed by India.

Table *IV* Top publishing countries 1994-2023

Country	**No. of publications**
US	97
Italy	25
Canada	22
China	13
Germany	11
Great Britain	11
Brazil	8
India	6
Saudi Arabia	6
France	4

Source: Table by authors

Figure III illustrates country co-authorships distribution. Closer countries on the map indicate stronger relatedness, with thicker lines denoting stronger links. The US led with 16 affiliations and 123 co-authorships, followed by Italy (8 links, 27 co-authorships), Canada (10 links, 24 co-authorships), and China (4 links, 15 co-authorships). These findings emphasize worldwide collaborations in the subject.

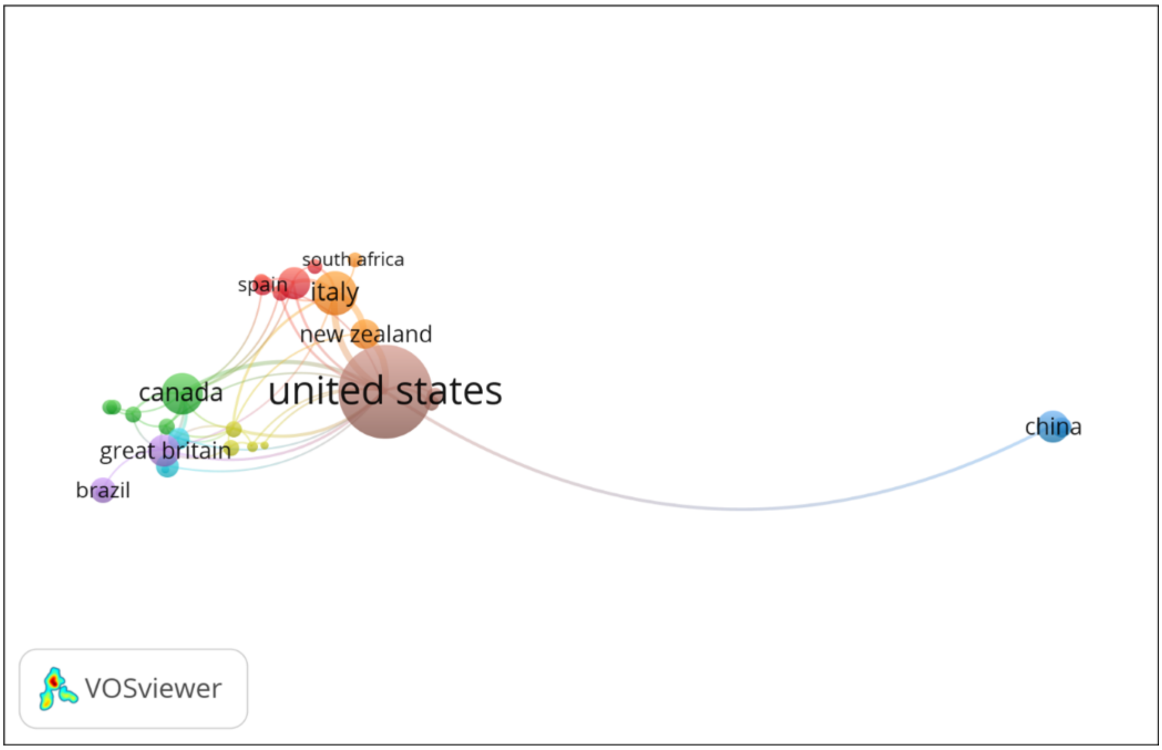

Figure III A screenshot of bibliometric map created based on co-authorship with network visualization mode (Link: http://tinyurl.com/ywuq44kt)

iii. Keyword analysis

We examined keyword co-occurrence, identifying 942 terms, with zero occurrence. These are based on author-provided keywords, publication titles, abstracts, and full texts (Donthu *et al.*, 2021). Keyword analysis uncovers popular research areas. Table *V* reveals "assistive technology" as the most prevalent keyword, with 138 occurrences, reflecting its importance in workplace accessibility and inclusion.

Table *V* Authors' Top 10 keywords

Keywords	**Occurrences***
Assistive technology	138
Disability	41

Brain-computer interface (BCI)	28
Human-computer interaction (HCI)	20
Rehabilitation	17
Visual impairment	16
Employment	14
Accommodations	14
Computer access	12
Electroencephalography (EEG)	11

Source: Table by authors

Figure IV displays keyword connections within clusters. The strongest link (strength 18) is between "assistive technology" and "disability," followed by "assistive technology" and "brain-computer interface" (strength 17). This reveals that these terms received the most research attention.

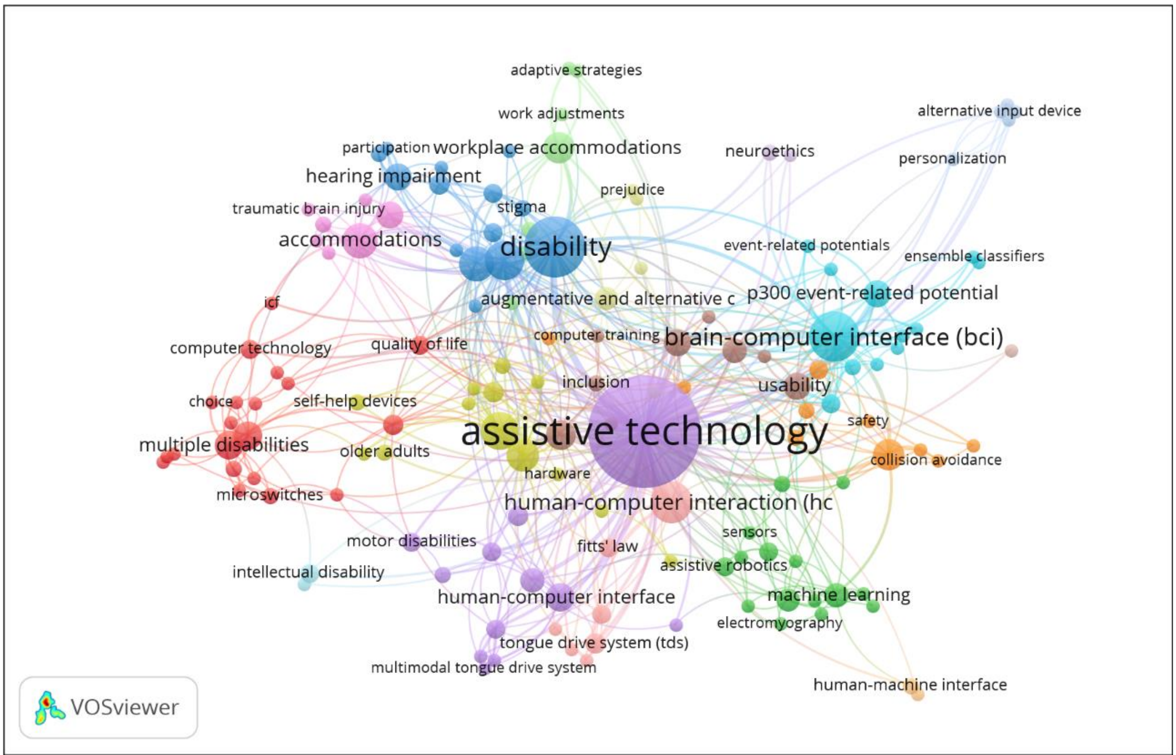

Figure IV A screenshot of co-occurrence network of author keywords (Link: http://tinyurl.com/yrwzeb29)

The VOSviewer identified 17 keyword clusters representing research emphasis. Table *VI* lists the top three clusters. To make further sense, we examined details of these clusters.

Cluster 5 encompasses keywords around AT, like eye tracking and speech recognition designed for cerebral palsy and motor disabilities, capturing a range of technologies for people with physical and communication challenges. Cluster 3 explores topics such as disability rights, workplace discrimination and rehabilitation, touching on the Americans with Disabilities Act (ADA) compliance, workplace inclusivity, and discrimination. Cluster 6 focuses on cutting-edge technology including BCI, virtual reality, and cloud computing, emphasizing user-centered design for tailored technology solutions.

These clusters illustrate technology roles in empowering PWD at work through assistive tools; addressing broader disability rights issues and providing personalized technology solutions.

Table *VI*. Top three clusters

Rank	**Cluster**	**Keywords**
1	Cluster 5	Assistive technology; Cerebral palsy; Cursor control; Eye tracking; Gaze estimation; Head tracking; Human-computer interface; Motor disabilities; Multimodal tongue drive system; Proportional control; Speech recognition; Tetraplegia
2	Cluster 3	ADA; Disability; Discrimination; Employment; Hearing impairment; Mobile device; Occupational health; Participation; Qualitative analysis; Rehabilitation; Stigma; Work; Workplace; Workplace discrimination
3	Cluster 6	Amyotrophic lateral sclerosis; BCI; Brain-machine interface; Cloud computing; Ensemble classifiers; Event-related potentials; N-class

	classification; P300 event-related potential; SSVEP; Steady-state visual evoked potential; User-centered design; Virtual reality

**Contact the authors for the full list.*

Source: Table by authors

3.2 Systematic Review

We conducted SR following PRISMA guidelines. This process identified 114 articles relevant to AT usability for PWD at work.

Three themes and several sub-themes emerged focusing on success factors for technology empowerment in PWD employment/work settings. The themes are framed within the Theory of Work Adjustment (TWA) for individual and environment factors, and the Social Model of Disability theory for societal factors. Individual factors refer to characteristics, abilities, and needs of PWD. Environment factors represent workplace elements influencing PWD integration. Societal factors encompass broader influences on PWD integration. These themes shape PWD's AT experiences. Table *VII* shows sampled articles.

Table *VII* Summary of Sample Papers

Themes	**Sub-themes**	**Sample articles**	**Technologies**	**Study focus**	**Disability**	**Summaries**
Individual factors	Influence of PWD participation, independence, and personal attributes	(Burgstahler *et al.*, 2011) (Dong *et al.*, 2021) (Collinger *et al.*, 2013)	Computers, mobiles & other ATs, BCI	Participation, independence, resilience and self-advocacy	Neurological / mobility impairment, visual impairment, spinal cord injury	These attributes help PWD overcoming obstacles and use technology to improve independence and performance.
	Influence of physical and cognitive abilities / functional limitation	(Sahadat *et al.*, 2018) (Andreas *et al.*, 2022)	Head-tracking, speech recognition, tongue motion, computer	Cognitive assessment and functional capacity	Physical disabilities, people with neuromuscular diseases	AT usage can be understood by assessing cognitive and functional abilities.
	Influence of technology literacy	(Chi *et al.*, 2012) (Gupta *et al.*, 2021)	Computer, AT devices including alphanumeric entry, pointing, output, performance enhancement	Technology proficiency and AT familiarity	Visual impairment, PWD in general	PWD must be tech-savvy and conversant to use AT professionally.
	Influence of adaptive behaviour	(Lancioni *et al.*, 2012) (Gupta *et al.*, 2021)	Computers, microswitches, voice output communication devices	Adaptive behaviours and attitudes towards technology	Motor disabilities, PWD in general	Technology attitudes influence AT adoption.
	Influence of PWD educational background	(Yang, 2003) (Yeager *et al.*, 2006)	Computer, software and hardware devices, switches, augmentative and communication, mobiles, magnifiers	Educational background and skills for AT utilization	Physical disability, PWD in general	PWD gain AT skills through education, enhancing work prospects and overcoming employment barriers with this empowerment.
	Influence of motivation and confidence	(Lancioni *et al.*, 2012)	Special text messaging system, net-book	Adaptation on writing technology to	Multiple disabilities,	Overcoming hurdles and adapting to new

	towards technology	(Dong *et al.*, 2017)	computer, mobile communication modem, input microswitch, keyboards with Braille	the participants' characteristics	visual impairment	technologies requires motivation and confidence in learning and applying AT.
Environment factors	Workplace infrastructure and workspace design	(Gupta *et al.*, 2021) (Rouillard and Vannobel, 2023)	ATs, human-machine and human-robot interactions, computers with specialized software, smartphone, tablets, communication aid, workstation ergonomics	Accessibility features, ergonomic design, lighting, noise levels, use of voice, touch and gaze tracking	Visual impairment, PWD in general	Safe workplaces require accessible infrastructure and ergonomic design to reduce accident risks and enable PWD workforce integration.
	Technological infrastructure	(Gupta *et al.*, 2021) (Karjalainen *et al.*, 2022) (Betke *et al.*, 2002)	Hardware and software, computer, eye-tracking system	Tailored AT to the needs and requirements of PWD, workplace support systems, job-person fit and modifications to work environments and visual tracking.	PWD in general, cognitive impairments	Accessible technology and IT support enhance PWD work engagement.
	Organizational policies and practices	(Gupta *et al.*, 2021) (Dettmann and Hasselhorn, 2021)	Hardware and software, computer	Tailored AT to the needs and requirements of PWD, workplace support systems, job-person fit and modifications to work environments	PWD in general, health impairment	Clear organizational policies and support systems help PWD use AT and other workplace accommodations.
	Social support and attitudes	(Steverson and	ATs and computer	Social interactions, work supports	Visual impairment, cancer survivor	The importance of workplace social factors includes

		Crudden, 2023) (Baldridge and Swift, 2013) (Taskila and Lindbohm, 2007)		and social support		employer and coworker attitudes toward disability, job supports, and disability-friendly workplace cultures.
Societal factors	Stigma and discrimination	(Graham *et al.*, 2019) (Schreuer and Dorot, 2017)	ATs and computer	Societal stigma and prejudice	Physical, behavioural, neurological, and sensory impairments, hyperactivity disorder (ADHD)	Improving AT accessibility and PWD workforce inclusion requires addressing social stigma and prejudice.
	Policy and legislation	(Govender *et al.*, 2021) (Sabatello, 2014)	ATs and computer	Disability rights legislation, accessibility standards and Job Accommodation Network's (JAN) approach	PWD in general	Government policies, laws, and regulations promote AT accessibility and PWD workforce inclusion.

**Contact the authors for complete list.*

Source: Table by authors

Studies by Andreas et al. (2022), Chi et al. (2012), Collinger et al. (2013), Dong et al. (2021), Gupta et al. (2021) and Sahadat et al. (2018) explored individual factors like personal resilience, self-advocacy skills, independence, and tech familiarity in PWD. They examined how technology, including computers, mobile devices, and BCI, improves engagement and functionality for different disabilities. They found that successful AT application depends on factors like tech literacy and personal resilience. Betke et al. (2002), Dettmann and Hasselhorn (2021), Rouillard and Vannobel (2023) and Taskila and Lindbohm (2007) emphasized workplace infrastructure, policies, and social support to enhance PWD's

AT usage. They discussed technologies like specialized software, smartphones, and aids for accessibility and ergonomics, reducing risks and promoting integration. Clear organizational policies are essential for AT usability at work. Govender et al. (2021) and Graham et al. (2019) discussed societal stigma, discrimination, and government policies affecting AT accessibility for PWD. Overcoming social barriers and regulations are key to improving AT accessibility and inclusivity in the workplace.

These studies collectively underscore how individual, environmental, and societal factors interconnected in PWD's work performance. They highlight the need for holistic and inclusive technology practices to create a more welcoming work environment.

4. DISCUSSION

The study integrates insights from BA and SR findings to provide a comprehensive view of technology empowerment in AT for disability employment.

Our bibliometric findings highlight several points. Over the past three decades, AT research has shown consistent growth, demonstrating increased importance of PWD inclusion (ILO, 2023). Several Western and European countries, especially the US, dominate publications in the area. The depiction of research affiliations highlights strong collaboration, especially between the US and Italy, demonstrating a global effort to advance PWD technology. Keyword analysis revealed seventeen research concentrations on the subject. Keywords like "assistive technology," "disability," and "brain-computer interface" show continuous interests on cutting-edge technologies for impaired individuals. Research on "human-computer interaction" reflects a focus on user-centered design and technology accessibility for PWD. These findings highlight present and future interests on untapped areas for research.

Our SR highlights key factors influencing AT usage by PWD at work, categorize into individual, environment, and societal factors, aligned with the TWA and the Social Model of Disability. Our findings emphasize the intricate interplay of individual, environment, and societal factors in shaping the usage of AT by PWD in the workplace. Individual factors such as personality traits and technology acceptance influence the adoption of customized AT solutions.

The importance of workplace policies, infrastructure, and employer support is highlighted in the environment factors, while societal factors like stigma and discrimination significantly impact PWD integration in the workforce. To promote a supportive work environment, creating inclusive policies and initiatives that reduce stigma are imperative for empowering PWD and enhancing their participation at work.

Our analysis also highlights various examples of AT applications that have shown promise in improving engagement, independence, and resilience among PWD. These technologies, ranging from basic computer tools to cutting-edge solutions like BCI and eye-tracking systems, demonstrate the potential and impact of AT in enhancing workplace inclusiveness and accessibility. Thus, they provide actionable knowledge for future research and practice.

Our broad-based approach treats PWD as one group, yet it underlines the different AT demands of different disabilities. For instance, visual-impaired individuals often benefit from screen readers and Braille displays, while mobility-impaired people find adapted keyboards and switch controllers helpful. Additionally, eye-tracking systems can enhance engagement for those with cognitive impairments.
Our analysis supports the idea that AT tools are crucial for PWD, helping overcome employment barriers and fostering inclusivity and engagement at work. Using the right tools ensures that AT solutions meet specific needs; making workplaces more accessible for everyone.

Based on the findings, we propose a conceptual model of work adjustment and usability of AT (Figure V), built upon the TWA (Dawis *et al.*, 1964), HCI Theory (Carroll, 1997), and the Social Model of

Disability (Shakespeare, 1996). TWA provides a dynamic understanding of how individuals adapt to their work environment, while HCI sheds light on the usability and user experience of AT.

The Social Model of Disability Theory emphasizes societal factors in creating barriers, prompting consideration of broader influences in AT research. The model highlights how various factors shape technology empowerment in disability employment. It reflects the need to adopt multiple theoretical perspectives to better support PWD in creating more inclusive workplace and accessible society for everyone.

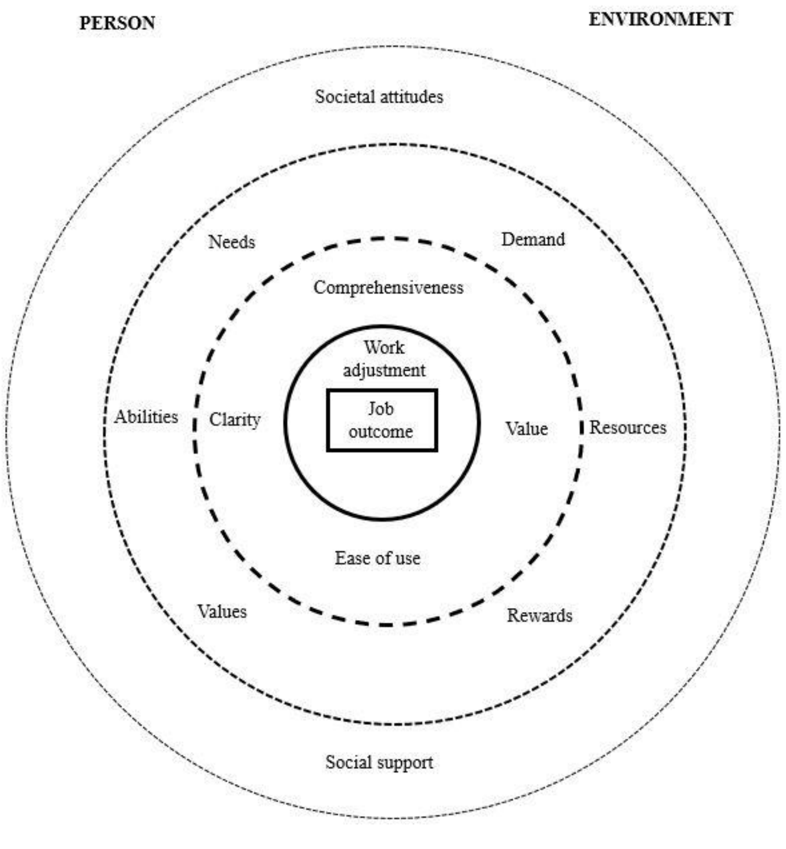

Figure V The conceptual model of work adjustment and usability of AT

Source: Authors’ development

Although our study was primarily interested in PWD employment/work settings, the findings transcend to other spheres for PWD's overall social well-being. The interplay between individual-environment-societal factors in our model is applicable for PWD social inclusion in general. Labor market integration of disabled people requires support from stakeholders involving multi-level policy interventions which include healthcare, economic, social, education and human capital (Hogelund, 2003).

This integration can significantly enhance support systems, improve access to services, and ultimately contribute to a more inclusive and supportive society for PWD communities. Having jobs serve more than economic purpose but is pertinent for individuals' overall quality of life (Parola and Marcionetti, 2022; Pixley, 2018). For PWD, employment is a human rights issue, enabling them to fully participate and be a productive citizen.

AT provides the tools that can empower and ensure social and welfare of PWD. Aligned with this purpose, our research adds understanding of technology empowerment through AT usage in disability employment and broader societal implications.

5. CONCLUSION

Our study had four objectives. Firstly, we analysed WoS and Scopus to map research articles published between 1994-2023. Surging publications post-2008 suggest a growing focus on technology solutions for disabilities. While there has been significant progress, some context requires further exploration. For instance, articles on Southeast Asian are notably lacking, pointing to research needed in the region. The second objective was achieved by producing the key terms and prevailing research themes in the area. The keyword analysis highlights the crucial role of technology, encompassing common AT tools as well as advanced systems like AI, robotics, and machine learning, which empower PWD in the workplace. Given the evolving nature of technology and its applications in the workplace, ongoing research efforts are necessary to investigate the capabilities and suitability of tools across various disabilities and work settings. For the third objective, based on SR, the study identified three key themes: individual factors, environment factors, and societal factors. These shed light on the multi-faceted influences shaping technology utilization by PWD in workplace settings. The fourth objective proposes a conceptual model grounded in the TWA, HCI theory, and the Social Model of Disability. This model emphasizes a holistic perspective encompassing individual, environment, and societal factors in fostering technology empowerment for disability employment. It incorporated theoretical perspectives from psychology, human resources, human technology, and social science disciplines to capture the complexity of disability inclusion that requires multipronged strategies.

Future research can adopt a similar multi-disciplinary stance in researching PWD employment. Furthermore, a notable scarcity of studies in social science, human resources, and management contexts necessitate further exploration. More studies are needed that can simultaneously merge inputs from key stakeholders in PWD employment. This relates to the lifecycle of human capital development involving disabled individuals, rehabilitation / vocational services, employers, policy makers, and social services.

Our study raises several points for policymakers and practitioners. Governments need to put more priorities to encourage technology empowerment among disabled individuals. These can include budget allocation to purchase devices, grants for employers to embed technology for work accommodation, research, and development to develop appropriate tools, and technology-based training for practitioners involved in disability social services. Technology can be expensive, hence depriving certain groups from its benefits. Therefore, collaboration between stakeholders can aid the financing process.

The UN Sustainable Development Goals (SDG) that list equality and decent work as its goals can be used as an important platform to encourage countries to enhance disability rights. Digitalization / AI agenda, developed by many countries, can also push for wider application of social technologies for PWD. Exposure to technology can be done at a much earlier age to enable PWD empowerment. As mentioned, socio-economic inclusion of disabled people cannot be approached in isolation but needs holistic interventions.

Our study contains limitations. Although we have covered two major databases, we may have missed some important publications which appeared elsewhere. Future review studies may explore other databases to expand the understanding. We purposely focused on research articles to suit the study objectives.

Future researchers can include other types of publications such as book chapters and non-English documents to widen the data capture. Alternative search keywords, indicating different forms of AT and disabilities, can be employed to enhance result specificity. Such focused analysis can provide meaningful insights on AT application associated with different disabilities to allow targeted employment interventions for PWD.

In conclusion, this study marks a step towards understand technology empowerment for disability employment. It offers a foundation for researchers, policymakers, and practitioners to collaborate in empowering the PWD communities.

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[1] The remaining 142 articles were excluded for SR because they focused on non-work-related topics, i.e. technology assessment in labs (117 articles), rehabilitation (6), clinical settings (11), and too general issues (8).